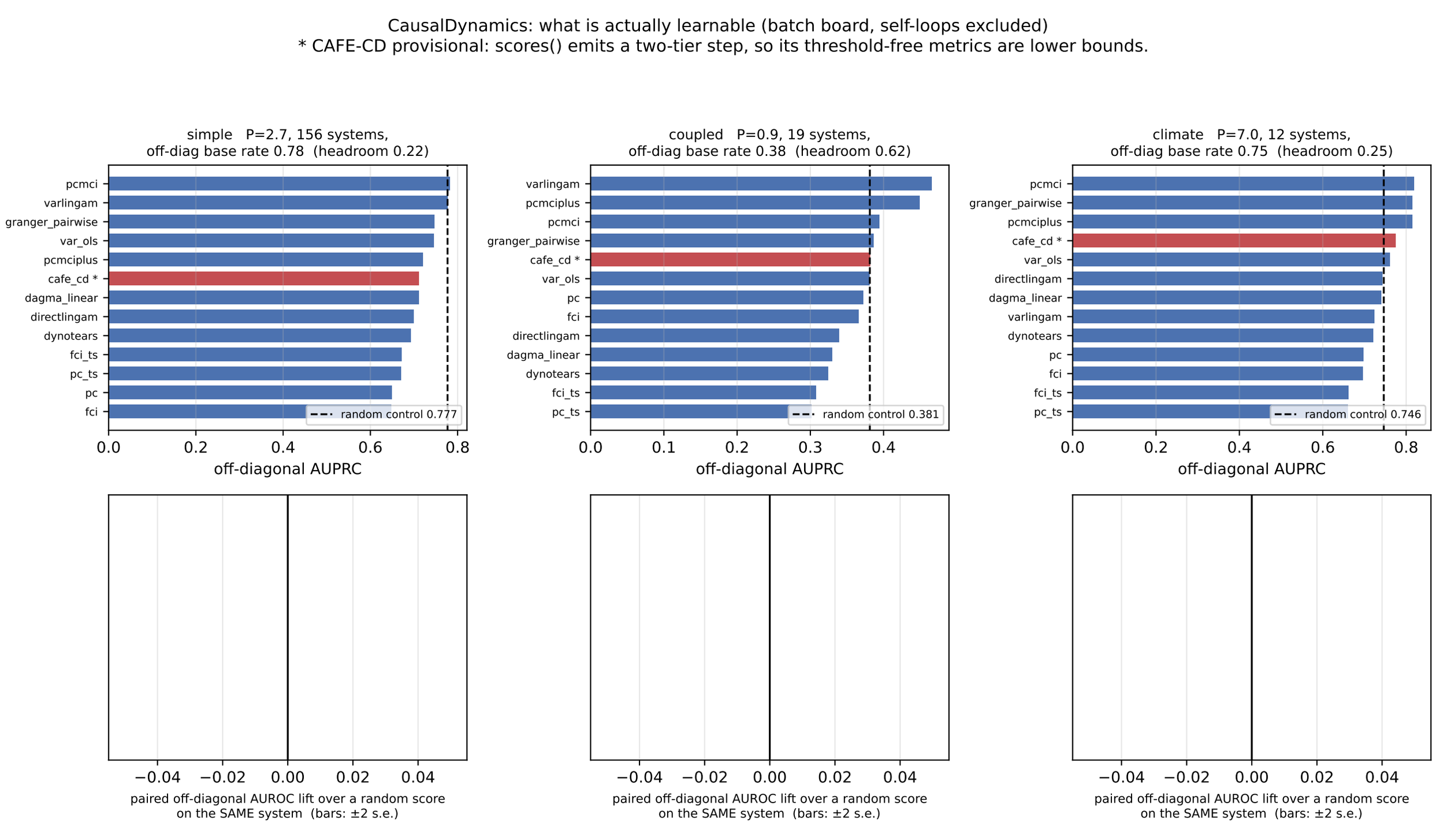
CausalDynamics: what is actually learnable (batch board, self-loops excluded)
* CAFE-CD provisional: scores() emits a two-tier step, so its threshold-free metrics are lower bounds.
simple P=2.7, 156 systems,
off-diag base rate 0.78 (headroom 0.22)
pcmci
varlingam
granger_pairwise
var_ols
pcmciplus
cafe_cd *
dagma_linear
directlingam
dynotears
fci_ts
pc_ts
pc
fci
random control 0.777
0.0
0.2
0.4
0.6
0.8
off-diagonal AUPRC
coupled P=0.9, 19 systems,
off-diag base rate 0.38 (headroom 0.62)
varlingam
pcmciplus
pcmci
granger_pairwise
cafe_cd *
var_ols
pc
fci
directlingam
dagma_linear
dynotears
fci_ts
pc_ts
random control 0.381
0.0
0.1
0.2
0.3
0.4
off-diagonal AUPRC
climate P=7.0, 12 systems,
off-diag base rate 0.75 (headroom 0.25)
pcmci
granger_pairwise
pcmciplus
cafe_cd *
var_ols
directlingam
dagma_linear
varlingam
dynotears
pc
fci
fci_ts
pc_ts
random control 0.746
0.0
0.2
0.4
0.6
0.8
off-diagonal AUPRC
−0.04
−0.02
0.00
0.02
0.04
paired off-diagonal AUROC lift over a random score
on the SAME system (bars: ±2 s.e.)
−0.04
−0.02
0.00
0.02
0.04
paired off-diagonal AUROC lift over a random score
on the SAME system (bars: ±2 s.e.)
−0.04
−0.02
0.00
0.02
0.04
paired off-diagonal AUROC lift over a random score
on the SAME system (bars: ±2 s.e.)